

Research article

Femtosecond laser assisted, non-toxic and complete delamination of photovoltaic module layers

Radhesh Sanathi^{a,b}, Shantanu Bhowmik^b, Sourish Banerjee^{a,*}^a School of Engineering, University of Southern Queensland, Toowoomba, 4350, Queensland, Australia^b Department of Aerospace Engineering, Amrita Vishwa Vidyapeetham, Coimbatore, 641112, Tamil Nadu, India

ARTICLE INFO

Keywords:

Femtosecond laser
Ethylene vinyl acetate
Delamination
Photovoltaic module
Recycling
Solar panels

ABSTRACT

Recycling solar panels is key to curbing growing photovoltaic (PV) waste, and separating different material layers is the critical step in the process. The current delamination methods mainly utilize toxic chemicals for dissolving EVA or decomposing EVA through pyrolysis, which produces harmful gases. In this article, a femto-second laser delamination treatment is proposed to separate PV module layers. Experimental results show that laser irradiation of module front with both 1030 nm (IR) and 515 nm (green) lasers results in clean separation of front EVA from both cell surface and glass, demonstrating effective separation at each interface. This laser treatment was performed twice, first on the back to separate the back EVA from the cell and then from front, ultimately separating all the layers of the module. This method shows great potential for the non-toxic delamination of PV modules. The laser energy deposited onto the cell modifies its surface, which in turn breaks the cell-EVA interfacial bond. The nonlinear nature of the interaction between ultrafast femtosecond laser and glass-EVA interfacial bond, along with heat transmitted by the laser during transmission are most likely causes for glass-EVA separation. Separated EVA is tested for damage through FTIR, TGA, and EDS characterisation techniques. In addition, for the effective and undamaged separation of EVA layer, a parametric study is undertaken to determine the optimized values of fluence and pulse repetition rate. It is also observed that single laser treatment from the front can separate glass, front EVA, and backsheet.

1. Introduction

Owing to its sustainability, scalability, low operation costs and abundance of sunlight, solar energy has become one of the leading renewable energy sources. This has led to global solar energy generation surpassing 2000 TWh in 2024, which is equivalent to 7% of global electricity generation, and by 2027, it is expected to generate half of increase in global electricity demand (IEA, 2025). With the lifespan of a typical crystalline silicon solar panel being 30 years, it is estimated that by 2050, 78–88 million tonnes of waste solar panels will be accumulated globally (Al Zaabi and Ghosh, 2024; IEA-PVPS, 2016; Keiichi Komoto et al., 2025). Disposal of end-of-life solar panels is difficult owing to their complex laminated structure and the presence of heavy metals. In addition, to deal with future metal scarcity and to promote resource circularity, it has become essential to recycle end-of-life solar panels (Hallam et al., 2023; Jain et al., 2022; Kwak et al., 2020; Meena et al., 2024; Ngagoum Ndalloka et al., 2024). Typical crystalline silicon photovoltaic (PV) modules are surrounded by an aluminium outer

frame. The top layer of module is tempered glass (thickness ~ 3.2 mm), the solar cells below are encapsulated by Ethylene-vinyl Acetate (EVA) polymer (thickness ~ 0.46 mm) and a protective backsheet (thickness ~ 0.34 mm) at the bottom (Dobra et al., 2023) (Fig. 1). To extend the outdoor durability of the panel, additives like curing agents, photo-thermal antioxidants, and UV absorbers (Agroui et al., 2006), rare earth fillers like $\text{Y}_2\text{SiO}_5\text{:Ce}^{3+}$, Yb^{3+} (Xue et al., 2014) are added to EVA encapsulant. Plasma surface treatment of the cell increases the surface energy, thereby increasing the bond strength with the encapsulant (Technology, 2025). These additives and treatments increase the lifespan of the solar panel by enhancing its durability but also complicate the recycling process. All the layers are laminated together to produce a single panel. The interfacial adhesion between EVA and glass could be achieved by molecular penetration or chemical bonding (Si-O-Si bond at interface) (Koenig et al., 1986; Meena et al., 2024). To recycle a solar panel, the primary challenge is separating each layer, which is the main aim of this work.

The current delamination technologies can be broadly divided into

* Corresponding author.

E-mail addresses: u1166260@umail.usq.edu.au (R. Sanathi), b_shantanu@cb.amrita.edu (S. Bhowmik), Sourish.Banerjee@unisq.edu.au (S. Banerjee).

physical, thermal, and chemical techniques. *Physical techniques*, including crushing (Fiandra et al., 2023; Granata et al., 2014; Pagnanelli et al., 2017), high-voltage crushing (Song et al., 2020), scraping (Fiandra et al., 2023), and sieving (Fiandra et al., 2023; Granata et al., 2014; Pagnanelli et al., 2017; Song et al., 2020) have good throughputs but still yield polymer-rich fractions that need further thermal treatment to remove EVA stuck to the crushed fractions. *Thermal techniques* involve decomposition of polymer layers (Fiandra et al., 2019; Fiandra et al., 2019; Lee et al., 2018; Xu et al., 2021). Although these techniques offer complete separation, they also emit toxic gases. *Chemical techniques* involve dissolving polymer layers using various solvents like trichloroethylene (Doi et al., 2001), O-dichlorobenzene (Kim and Lee, 2012), hexane (Tembo et al., 2021), and KOH-ethanol (Yan et al., 2020). They can also provide effective separation, but majority rely on hazardous chemicals that need treatment after use. The adoption of green solvents like D-Limonene (Abdo et al., 2023), DMPU (N, N'-dimethylpropenylurea) (K. Li et al., 2022), EGDA (ethylene glycol diacetate) (Min et al., 2023) and ChCl-Oxa Deep-Eutectic solvent (Yu et al., 2024) can mitigate this issue, but the process remains inherently time-consuming because the solvent penetration for the entire panel takes time. For effective delamination, innovative techniques like hot knife cutting (Ito and Doi, 2024; Wahman et al., 2024) and laser delamination (Bin Anwar et al., 2024; X. Li et al., 2022) were also explored. Hot knife cutting is straightforward and effective, but EVA still adheres to the solar cells, requiring further treatment. The current laser delamination techniques were shown to be effective in separating back EVA and front EVA from the cell, but haven't addressed the separation of glass EVA interface, requiring an additional step. As delamination is the most critical challenge of PV recycling, it is necessary to achieve clean separation of all the layers in a non-toxic manner to develop a much-needed sustainable process.

This study aims to separate the glass, front EVA and cell of the solar panel using femtosecond laser irradiation. It addresses the issue of complete separation of all the layers in a nontoxic manner. The quality of separation and the extent of damage to the separated layers are assessed. A parametric study is also undertaken to determine suitable laser parameter values for which the separation is clean and the separated front EVA layer is undamaged. This contrasts with previous works that used nanosecond laser to achieve separation at only one interface, either front EVA/cell (Bin Anwar et al., 2024) or cell/back EVA (X. Li et al., 2022). Importantly, this approach simplifies the separation

process by eliminating the need for additional treatment step to separate glass and front EVA.

For clean separation, it is necessary to break the interface bond between different layers. As glass and EVA are transparent, Fig. 1(b) and (c) show that the laser is focused on the top surface of the solar cell, depositing its energy at the front EVA/cell interface to break it. The energy from laser is absorbed by the top surface of the cell, localized heating of the cell breaks the bond between cell and front EVA. Since we don't want to destroy the rest of the structure and only affect the targeted zone, pulsed laser was used instead of a continuous laser. A pulsed laser, contrary to a continuous laser, deposits energy in a short interval of time, and the gap between each pulse also prevents the material from building up too much heat. Hence, in this study, a femtosecond ultra-fast laser was used instead of a longer pulse-width nanosecond laser (Bin Anwar et al., 2024) as it limits the heat-affected zone and thus, causes localized damage (Brand et al., 2023; Chris et al., 2001; Spectroscopy, 2013; Weber et al., 2014). The interaction between incident femtosecond laser and the surface of the irradiated material is nonlinear and damage mechanism mainly includes avalanche ionization, multiphoton ionization and thermal damage (Guo et al., 2019; Lenzner et al., 1998; Pan et al., 2023). The following subsection provides an overview of the physical mechanisms underlying the laser-material interactions.

1.1. Theoretical framework

This section introduces the fundamental principles underlying the laser-material interactions relevant to this work. Laser material interaction starts with photons in laser being absorbed by the material. Laser modifying the properties of optical medium in nonlinear manner is referred to as nonlinear absorption. Ultrafast lasers like femtosecond laser induce nonlinear absorption phenomena like avalanche ionization and multiphoton ionization when absorbed by the material. The electric field of ultrafast lasers femtosecond laser accelerates the free electrons in the material which generates more free electrons by colliding with neutral atoms. This process leads to exponential growth of free electron density. This phenomenon is called avalanche ionization (Wright et al., 2019). Multiphoton ionization (MPI) occurs when valence electron simultaneously absorbs multiple low energy photons exceeding its ionization potential. This phenomenon lets the laser interact with material which is typically transparent to the laser of this wavelength (Sharma et al., 2018; Yang et al., 2025). These nonlinear absorption

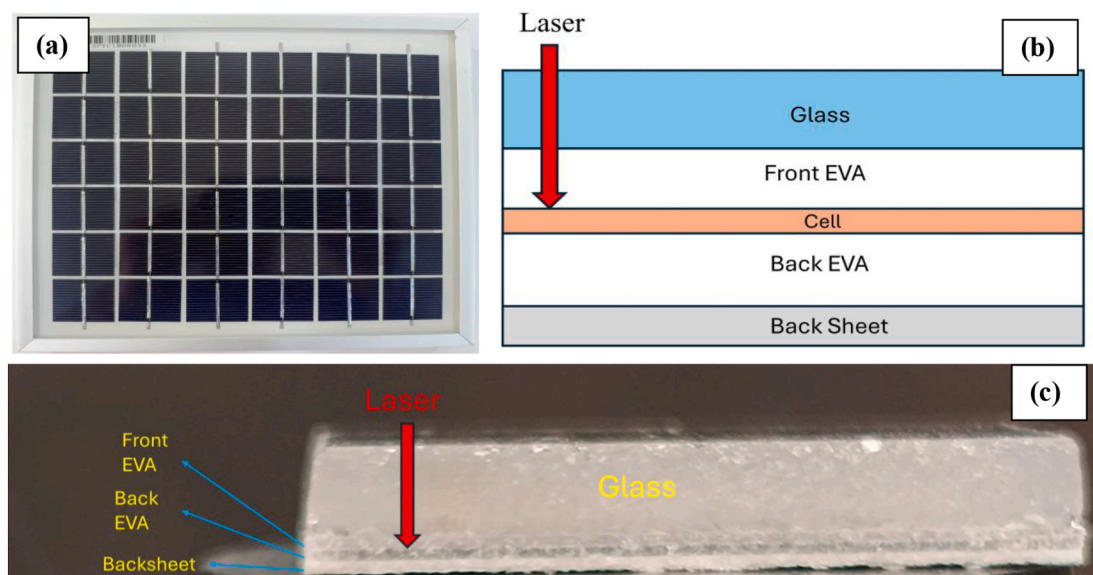


Fig. 1. (a) a Crystalline silicon solar panel (b) Schematic diagram of the solar panel, (c) Cross-sectional image of a real photovoltaic module, showing laser focused on the top surface of the solar cell (cell not visible because the sample was not cut through the cell).

phenomena induce plasma formation further leading to laser ablation, which can cause delamination between layers. Glass exposed to high intensity ultrafast femtosecond laser could lead to the above-mentioned nonlinear phenomenon, even if the glass is transparent to the laser wavelength. This can induce the formation of color centers, which arise from crystal defects allowing additional wavelength of light to be absorbed in those centers (Chen et al., 2024). The formation of these color centers is the reason for photodarkening mentioned later in this article. As laser passes through the glass and EVA layers, it gets partially absorbed which in turn heats the material. This process is called thermal energy transmission which involves delivery of thermal energy to the material through a laser beam.

2. Experimental

2.1. Sample preparation

For the delamination experiments, a small commercial 5 W solar panel (205 mm × 230 mm × 18 mm) by Powertech from Jaycar was used. It is a 36-cell, multi-crystalline silicon solar panel weighing 0.64 kg with a voltage of 17.6 V and a current of 0.29 A at maximum power. The back junction box and aluminium frame were detached manually. The frame and junction box removal were performed carefully, not to crack or damage the solar panel. For the ease of experimentation, samples of the size of each solar cell (30 mm × 25 mm) were cut out from the panel using waterjet. This solar panel used hardened float glass instead of tempered glass, so the glass remained uncracked even after cutting the panel with waterjet. For some samples, the back sheet was heated using hot air gun and was manually peeled off, exposing the back EVA.

2.2. Laser characteristics and optimized parameters

For the laser irradiation treatment, a Light Conversion 40W Carbide CB3 laser (pulsed, femtosecond) was used with an Excelliscan 14 galvanometer and Optic Sil 163 mm F-theta optic lens fitted inside a Pulsar Photonics RDX800-laser chassis. The laser was operated at both its fundamental harmonic 1030 nm (IR) and second harmonic 515 nm (green). The authors did not have access to a 355 nm laser; 257 nm was available. However, due to photodarkening effect, 257 nm was excluded. At this low wavelength, most of the laser energy is absorbed by the glass, drastically reducing the separation process's efficiency. While 355 nm could be promising, significant photodarkening would be expected with this, as it was already observed with 515 nm laser. Given the unavailability and high possibility of photodarkening associated with 355 nm, it was not considered for this study. The laser spot sizes were determined to be 40 µm and 25 µm in diameter for 1030 and 515 nm, respectively. Laser parameters like power (P), pulse repetition rate (PRR), pulse width (PW) and pulse overlap were varied. A single-shot treatment was performed, with each spot on the sample irradiated by a single laser pulse. The samples were scanned using raster pattern, and for the ease of experiments, the overlap percentage of laser spots in both horizontal and vertical directions was kept identical. So, the scan speed depended on the pulse repetition rate and pulse overlap percentage. The scan speed in horizontal direction $S_h = (1 - \text{pulse overlap}) \times \text{spot size} \times \text{pulse repetition rate}$. So, PRR is directly proportional to scan speed, while pulse overlap is inversely proportional to scan speed. Laser fluence (F), which is energy delivered per unit area ($\text{J} \cdot \text{cm}^{-2}$) is used to assess the laser-induced damage to the material.

Firstly, the laser irradiation experiments were conducted using IR (1030 nm) laser. Initially, a small area of 5 mm × 5 mm was scanned with different pulse widths (200 fs, 1000 fs, 5000 fs and 10,000 fs) to look for the effect it has on the sample. Once the right pulse width was identified based on its usability for the procedure, the samples were irradiated with high fluence values to assess the effectiveness of the proposed procedure in separating the layers. After the effectiveness was confirmed, fluence and PRR were optimized. Similarly, for 515 nm green

laser, correct pulse width was identified first, followed by the optimization of fluence and PRR values.

2.3. Analytical techniques used to characterize recovered materials

To assess damage to the cell and the separated EVA layer, scanning electron microscope (ZEISS Gemini 300) images and energy-dispersive X-ray spectroscopy readings were obtained. To assess changes in composition of the separated EVA, FTIR (Thermo Scientific iS50) and TGA (TA Instruments SDT650) experiments were performed.

3. Results and discussion

3.1. Delamination with IR laser: 1030 nm

Initially, the effect of different pulse widths at the interface of the irradiated area of the sample was observed to select the correct pulse width for the experiment. For this, four small squares of 5 mm × 5 mm area were irradiated with 200 fs, 1000 fs, 5000 fs and 10,000 fs pulse widths with the following parameters: PRR = 100 kHz, average power $P = 1.6 \text{ W}$, $F = 1.273 \text{ J} \cdot \text{cm}^{-2}$ and 60% laser spot overlap.

As shown in Fig. 2(a), the four squares, from left to right, represent the areas irradiated by the laser with different pulse widths, ranging from 200 fs on the left to 10,000 fs on the right. The change in the squares' colour and transparency can attest to the damage at the interface. The darker and more opaque the square appears, the greater the damage the interface sustained. The 200 fs square is very light and almost transparent, so least damage is done. The 1000 and 5000 fs squares are darker, but we can still slightly see the silver fingers on the cell through them. This indicates there is still contact at the interface. The 10,000 fs square is fully opaque, silver fingers are not seen and the EVA even took the shape of the pattern on the inner surface of the glass; both these details indicate separation. Light browning (photodarkening) of glass was observed for area irradiated with 200 fs. This means the laser's energy is absorbed by the glass and does not reach the cell's top surface, making the process inefficient. This explains why the 200 fs square appears very light, indicating low damage at the interface. Photodarkening is more pronounced with 515 nm laser, as discussed in the following subsection. For a 1030 nm laser, maximum damage at the interface occurred at 10,000 fs pulse width, so subsequent experiments were conducted using this pulse width. In order to determine if the femtosecond laser is effective in causing delamination, the sample shown in Fig. 3(a) with backsheet intact is irradiated with 10,000 fs pulse-width laser at 100 kHz PRR, 60% overlap and $3.183 \text{ J} \cdot \text{cm}^{-2}$ fluence. The laser is focused on the top surface of the solar cell, as shown in Fig. 1(b) and (c). The result, as shown in Fig. 3(c) and (d), is that the front EVA and cell are entirely separated. The laser can't pass through the edge of the glass, so the edges of the sample are still bonded. With a small amount of mechanical or peeling force, the edges can be separated, and the entire irradiated region of the cell and EVA gets detached without applying any force. However, because the edges were separated manually and the laser doesn't reach the edges, small fragments of cells remain adhered to the separated EVA along the edges. This edge separation is an inherent issue when working with small samples. However, it could become less significant if applied on full-size solar panel, as larger sample has lower edge-to-area ratio than a small sample. It was also observed that the front EVA layer could also be peeled off from the glass with small amount of force (Fig. 3(e) and (f)). So, by focusing the femtosecond laser on the top surface of the solar cell as shown in Fig. 1(b) and (c), two interfaces, namely front EVA/cell (Fig. 3(c) and (d)) and glass/front EVA (Fig. 3(e) and (f)), got separated. Visually, the separation was clean, and no residue remained on the glass. The nonlinear interaction of femtosecond laser and the interfacial bond, along with heat conducted by the layers when laser passes through them are probably what caused the bond to break and subsequently leading to separation. It is unlikely that the heat transferred to the glass EVA

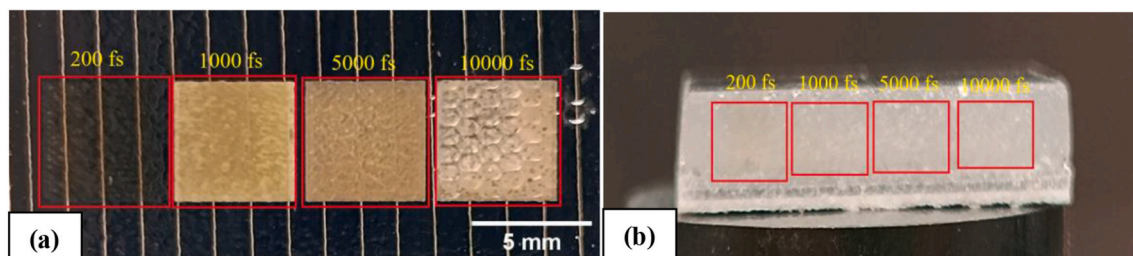


Fig. 2. (a) Four squares irradiated with different pulse widths, indicating damage at the interface and (b) corresponding light browning of glass due to photo-darkening on the left side of the sample.

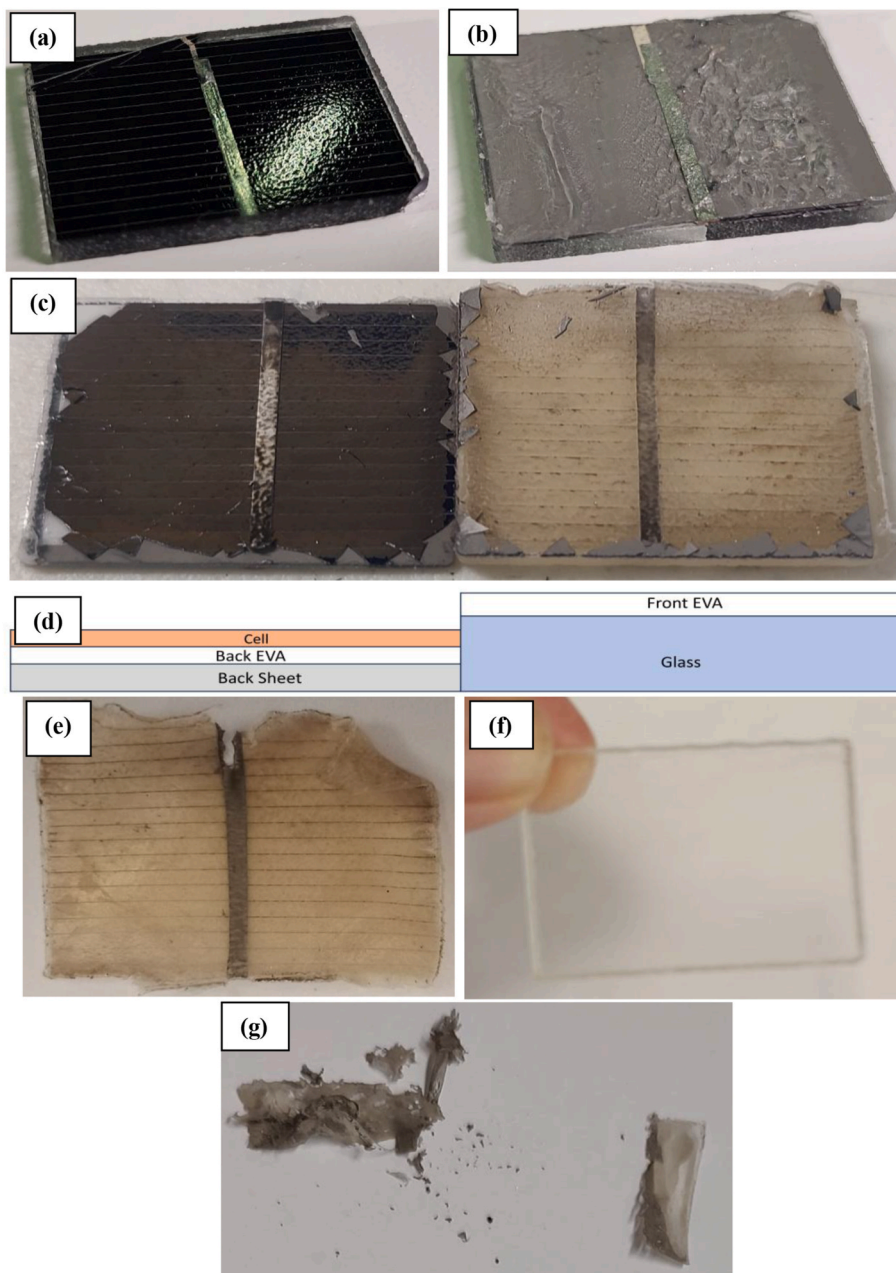


Fig. 3. (a) Front image of the sample, (b) Back image of the sample with backsheet removed, (c) Separation of EVA from the frontside of the cell, (d) schematic diagram for the split panel shown in c, (e) Separated front EVA, (f) Separated glass and (g) separated back EVA.

interface due to cell irradiation (i.e., heat transmitted from the cell after it absorbs laser energy) is the reason for separation, due to several

reasons. Firstly, EVA has poor thermal conductivity (Azizi et al., 2018), and secondly, if heat transferred were the cause, then the edges would

have been separated because of the heat transfer, but this was not observed.

Because the separation at the front EVA/cell interface was successful, it was expected that the same procedure would work at the back EVA/cell interface. To achieve complete separation of all the layers, the proposed process was tested for back EVA separation. For this, the back sheet of the sample was removed by peeling using hot air gun, the exposed back EVA can be seen in Fig. 3(b). First, the backside was irradiated with laser of same parameters, subsequently back EVA was removed as shown in Fig. 3(g). The back EVA was found to be uneven after peeling the backsheet and did not provide a good optical interface, so the laser caused more damage where the EVA was thin. Because of this, the back EVA could not be peeled as a single piece. As back EVA separation was not the focus of the study, it was not optimized, but the separation was still achieved using this laser. After removing the back EVA, the sample was irradiated from the front side, causing the separation of glass, front EVA and cell. Thus, separation of all the layers of the PV module was demonstrated to be possible with femtosecond laser in three steps, namely backsheet separation with hot air, back EVA separation with laser irradiation at the back and front laser irradiation to separate glass, front EVA and the cell. The separation of back EVA from the cell was previously achieved by Li et al. (X. Li et al., 2022) using a nanosecond laser.

The front side only laser treatment of the sample for these parameters took approximately 45 s for a surface area of 8.06 cm^2 , so normalized processing time is 5.58 s cm^{-2} . To quicken the process, PRR was increased to 500 kHz (simultaneously increasing the power to maintain similar pulse energy) and the rest of the parameters remained unchanged. The cell, front EVA and glass were effectively separated like the previous sample, but the process time reduced to approximately 9 s, reducing the normalized processing time to 1.11 s cm^{-2} . Double scan and laser burst setting were also tested to check for the effectiveness of the separation. Double scan refers to scanning the same area twice. This resulted in overheating, causing the EVA to char and leaving residuals on the glass. Burst mode splits a single pulse into multiple smaller pulses, all delivered within the same time span as the original pulse. Splitting single pulse into 3 smaller pulses lowers the energy supplied by each pulse. But front EVA was still charred and its residuals were observed on the glass. So, both double scanning and burst mode setting proved to be ineffective.

3.1.1. Assessment of the quality of separation and potential damage

3.1.1.1. Glass. To check the quality of separation between EVA and Glass (EVA residual on glass), the glass was heated to 550°C for 45 min in an oven. This will decompose any residual EVA (Dias et al., 2017). The glass was weighed before and after heating to calculate the weight of residual EVA. The initial glass mass was $\sim 6.5\text{g}$, 2 samples each were tested for both 100 kHz and 500 kHz. The difference in mass of the glass before and after heating (mass of residual EVA) lies between 0.2 mg and 0.9 mg. So, both the samples (100 kHz and 500 kHz) had less than 1 mg of residual EVA on them, indicating clean separation. As this residual EVA mass on glass is less than 0.25% of the mass of front EVA layer, this can be considered as clean separation.

3.1.1.2. EVA. To assess the damage to the separated front EVA, FTIR and TGA experiments were conducted on the EVA after laser irradiation at 100 kHz and 500 kHz repetition rates, both with 60% overlap and a 10,000 fs pulse width. Furthermore, SEM imaging of cell surface and selected area EDS analysis of EVA layer were performed to look for the damage.

Reference cell and reference EVA were compared with laser-separated cell and EVA. The reference EVA was peeled from sample, which didn't undergo laser treatment, while reference cell was collected by decomposing all the polymer layers (through pyrolysis (Dias et al.,

2017)) from the sample. In this way, both reference cell and reference EVA were collected from the sample without using laser. As can be seen in Fig. 4(a), the FTIR absorption spectrum of reference EVA, 2920 cm^{-1} and 2850 cm^{-1} peaks indicate stretching vibration of methylene (CH_2) group, while 1460 cm^{-1} and 720 cm^{-1} peaks indicate bending and rocking vibrations of the same group. 1730 cm^{-1} peak indicates stretching of carbonyl ($\text{C}=\text{O}$) group and 1370 cm^{-1} peak indicates bending vibration of methyl (CH_3) group. 1240 cm^{-1} and 1020 cm^{-1} peaks indicate antisymmetric and symmetric stretching of C-O bond respectively (Säckl et al., 2024). From Fig. 4(a), it can be seen that there is a broad peak between 3700 and 3100 cm^{-1} and small peak between 1700 and 1500 cm^{-1} in sample spectrum when compared to the reference EVA. The broad peak generally corresponds to hydroxyl region ($-\text{OH}$) and peaks between 1700 and 1500 cm^{-1} correspond to overlapping of $\text{C}=\text{C}$ bond and $\text{C}=\text{O}$ group region. These peaks represent the damage associated with photo-thermal degradation of EVA (Barretta et al., 2021). The $3700 - 3100 \text{ cm}^{-1}$ broad peak is associated with the formation of hydroperoxides from polymer degradation. These hydroperoxides could further decompose, forming carbonyl ($\text{C}=\text{O}$) groups (Uličná et al., 2023). The peaks in 1700 and 1500 cm^{-1} region due to $\text{C}=\text{C}$ could be the result of diffusion of additives to the surface followed by their decomposition (Oreski et al., 2009). Further optimization of the laser parameters will be performed to prevent the damage. TGA curves of EVA shown in Fig. 4(b) show that it degrades in two steps. First step involves the release of acetic acid at $\sim 350^\circ\text{C}$ and the second step involves degradation of the remaining hydrocarbons at $\sim 460^\circ\text{C}$, leading to complete decomposition (Agroui et al., 2016; Chitra et al., 2020). TGA curves of the samples trace that of reference EVA (Fig. 4(b)), indicating that the thermal bulk performance of the separated EVA remains unaltered despite the laser treatment. This apparent contradiction between unchanged bulk thermal properties (from TGA) and altered chemical structure (from FTIR) could be attributed to the degradation of additives. Additives are generally not homogeneously distributed, and when detected by FTIR in ATR mode, the measurements are only taken from the surface, as ATR mode has a limited penetration of few micrometers. Therefore, the chemical structure change detected by FTIR is only at the surface due to additives. Typically, individual additives constitute $<1\%$ mass of the total EVA polymer mass, so any chemical structural changes due to additives won't be represented in the bulk thermal properties measured by TGA.

3.1.1.3. Si-cell. The SEM image depicting the top surface topography of unirradiated cell (Fig. 4(c)) shows a typical honeycomb pattern of antireflective coating (SiNx) expected of a multi-crystalline silicon solar cell (Macku et al., 2012; Volk et al., 2015). The high fluence 3.183 J cm^{-2} used on the sample is well above the ablation threshold of silicon for a single pulse of laser with this wavelength and pulse width (Liu and Cheng, 2023; Sikora et al., 2017), even after considering some energy losses while transmitting through glass and EVA. So, the top surface of the cell with antireflective coating of thickness in the order of few tens of nanometres is fully destroyed (Fig. 4(d)) by laser ablation as the phenomenon is expected to have ablation depth of the same order (Kim, 2011; Liu and Cheng, 2023). This explains the grey appearance of the cell's top surface post-irradiation. SEM-EDS analysis was performed on the cell surface before (Fig. 4(c)) and after laser treatment (Fig. 4(d)). Before laser treatment, the surface consisted of 79.9% silicon and 15.3% nitrogen by weight. After laser treatment, the silicon content was 74.9% while nitrogen dropped significantly to 2.2% by weight. This, along with the destruction of surface topography, indicates that the anti-reflective coating may have been destroyed. The EVA layer may have detached due to the complete destruction of the interface bonded site, as the crosslinking of EVA at the interface is not reversible, the layers fail to re-bond after separation. The damage to the cell surface is not a serious concern for the proposed method, as the aim is to recycle the separated cell to extract metals as secondary source materials. Vitrimization

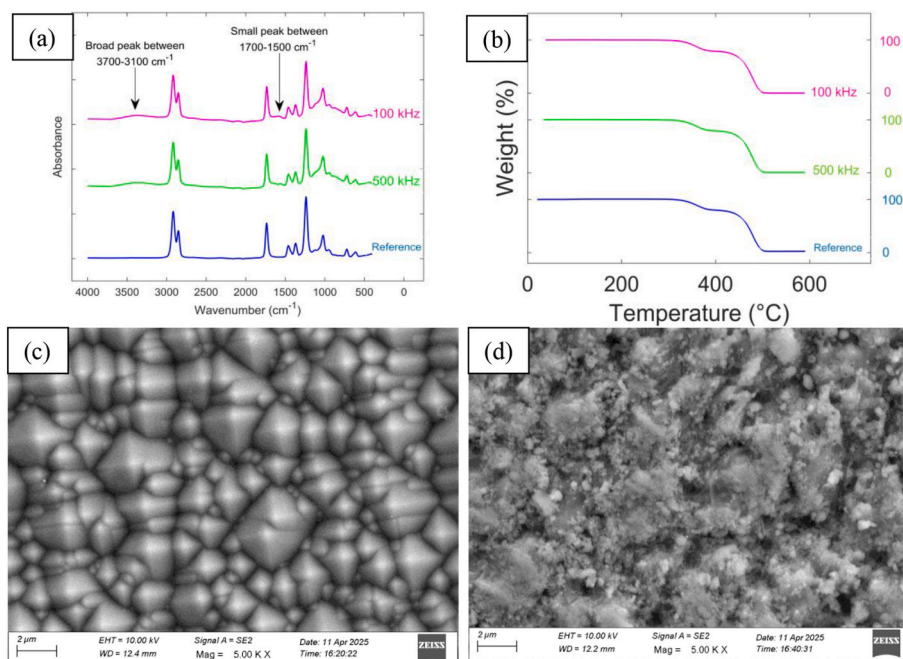


Fig. 4. (a) FTIR of separated EVA (100 kHz and 500 kHz samples), (b) TGA of separated EVA (100 kHz and 500 kHz samples), (c) SEM image of the top surface of a reference Si cell and (d) SEM image of the top surface of the sample Si cell irradiated with 100 kHz laser.

process proposed by (Bandegi et al., 2023) enables EVA to be remoulded, which could increase the potential for reuse of separated EVA in packaging materials and housing insulation applications (it requires further studies). It can also be downcycled and used as a filler in cement bricks. But separated EVA needs to be undamaged for the vitrimerization process to work. While the development of vitrimerization process is still in early stages and not at industrial application stage, the authors feel that it would be better to separate the EVA in undamaged state to enable any future recycling technologies. Therefore, additional experiments were conducted to optimize the fluence and PRR of laser to prevent damaging EVA layer while still achieving clean separation with glass and cell. Fluence and PRR were selected as the parameters because fluence controls the amount of energy deposited on a single spot and PRR controls the speed of the process, through which we can observe if delivering the same energy faster leads to EVA damage. The selected area EDS analysis of the EVA separated from the 100 kHz laser-treated sample (corresponding to Fig. 3 (d)) detected 53% of C, 20% of Pb, 11.8% of O, 11.3% of Sn, 3.7% of Si and 0.1% of Ag by weight. This shows that due to high fluence of the laser, ablation and phase explosion took place on the surface of the cell, leading to the ejected metal melt deposition onto the EVA.

To prevent damage to the EVA, fluence values lower than the previously used 3.183 J cm^{-2} were tested. To identify the range of fluence value to be tested, small squares (similar to Fig. 2) with the lowest fluence possible with this setup were irradiated on samples. From observing these squares, fluence of 1.047 J cm^{-2} seems to be the minimum necessary to cause damage at the interface. So, six fluence values 1.047 J cm^{-2} , 1.375 J cm^{-2} , 1.833 J cm^{-2} , 2.226 J cm^{-2} , 2.619 J cm^{-2} and 3.012 J cm^{-2} were selected for a systematic study. These values are between the minimum necessary to cause interface damage and the previously used (3.183 J cm^{-2}) high fluence value, which caused EVA damage. These six fluence values were tested for each repetition rate: 486 kHz, 243 kHz, 162 kHz, 97.2 kHz and 48.6 kHz. Throughout these experiments, pulse width and pulse overlap were kept constant at 10,000 fs and 60% respectively. For all these fluence and PRR values, clean separation was observed between cell and front EVA. So, the limiting factors are the separation of EVA from glass and damage to the EVA. TGA for these lower fluence samples were not performed because

there was no change in the separated EVA's curve when compared to reference EVA curve, even at high fluence.

Table 1 presents the FTIR results for the separated front EVA layer at the previously mentioned fluence and PRR values. In the table, undamaged EVA indicates that the FTIR spectrum of the sample shows no new peaks between 3700 and 3100 cm^{-1} or $1700 - 1500 \text{ cm}^{-1}$ and is similar to the spectrum of reference EVA. Damaged EVA has new peaks in the $3700 - 3100 \text{ cm}^{-1}$ or $1700 - 1500 \text{ cm}^{-1}$ range. While this method of using FTIR to check for damage can't estimate the extent of damage to the EVA, for the purpose of this work, it is sufficient to identify if EVA is damaged. The aim of this work is to separate the EVA to enable its potential reuse. So, parameter values that didn't cause any damage to EVA were selected as optimized values. The results indicate that there is no clear trend for undamaged EVA across the tested fluence levels and repetition rates. For all the tested fluencies at 243 kHz RRP, the EVA was damaged. While the damaged EVA in Table 1 shows the same peaks observed in Fig. 4(a), their intensities are much lower due to reduced fluencies. Samples irradiated with 1.047 J cm^{-2} fluence at 162 kHz and 48.6 kHz PRR showed no separation between the glass and EVA due to insufficient energy. As mentioned previously, while discussing the initial FTIR results, the damage could be due to degradation of additives and not the polymer itself, this could be the reason for the lack of good correlation between operating conditions and results.

While the interaction effects and underlying mechanism of the separation process are not fully understood within the parameter space, complete separation was achieved, and the process has proven to be reproducible, as the repetition of the experiment yielded consistent results. Further studies are required to enhance our understanding of the delamination and EVA damage mechanisms.

3.2. Delamination with green laser: 515 nm

Similar to the IR laser experiments, to select the correct pulse width for the experiment, the effect of different pulse widths at the interface of the irradiated area of the sample were observed. For this, four small squares of $5 \text{ mm} \times 5 \text{ mm}$ area were irradiated with 200 fs, 1000 fs, 5000 fs and 10,000 fs pulse widths with PRR = 100 kHz, average power of 1.12 W, $F = 2.285 \text{ J cm}^{-2}$ and 60% spot overlap.

Table 1Damage summary of the EVA separated by IR laser with fluence values lower than 3.183 J cm^{-2} for various repetition rates.

	PRR = 486 kHz	PRR = 243 kHz	PRR = 162 kHz	PRR = 97.2 kHz	PRR = 48.6 kHz
$F = 1.047 \text{ J cm}^{-2}$	Damaged EVA	Damaged EVA	No separation between glass and EVA	Undamaged EVA	No separation between glass and EVA
$F = 1.375 \text{ J cm}^{-2}$	Undamaged EVA	Damaged EVA	Damaged EVA	Undamaged EVA	Undamaged EVA
$F = 1.833 \text{ J cm}^{-2}$	Undamaged EVA	Damaged EVA	Damaged EVA	Undamaged EVA	Undamaged EVA
$F = 2.226 \text{ J cm}^{-2}$	Damaged EVA	Damaged EVA	Damaged EVA	Undamaged EVA	Undamaged EVA
$F = 2.619 \text{ J cm}^{-2}$	Damaged EVA	Damaged EVA	Undamaged EVA	Damaged EVA	Damaged EVA
$F = 3.012 \text{ J cm}^{-2}$	Damaged EVA	Damaged EVA	Undamaged EVA	Damaged EVA	Damaged EVA

As shown in Fig. 5(a), the four squares, from left to right, correspond to areas irradiated by the laser with different pulse widths, ranging from 200 fs on the left to 10,000 fs on the right. Here, the dark colour of the squares doesn't indicate damage at the interface; instead, it is due to the photodarkening of the glass. This is clearly visible in Fig. 5(b), for 200 fs square, most of the energy is absorbed by the glass, causing darkening and minimal damage to the interface. Though not so pronounced, 1000 fs and 5000 fs squares also caused photodarkening. Due to less darkening, there will be more energy deposited at the interface, but the efficiency of the process will still be low. For this reason, a pulse width of 10,000 fs was chosen for further experiments. Next, the effect of fluence (power) is observed on the sample at 10,000 fs pulse width, 100 kHz repetition rate and 60% overlap. For this, again five small squares were irradiated with fluence values of 0.083 J cm^{-2} , 0.355 J cm^{-2} , 0.796 J cm^{-2} , 1.453 J cm^{-2} and 2.323 J cm^{-2} while PRR and pulse overlap are constant at 100 kHz and 60% respectively. As shown in Fig. 5(c), the five squares from left to right correspond to areas irradiated by laser with different fluence values, ranging from 0.083 J cm^{-2} on the left to 2.323 J cm^{-2} on the right. From figure 0.083 J cm^{-2} seems too low a fluence (energy) to create significant damage to the interface, while rest of the fluence values seem workable. Photodarkening of glass was not observed for any of these fluence values. The same set of squares were irradiated for the same fluence values but for a pulse width of 5000 fs. In this case, 0.083 J cm^{-2} again was too low a value to cause interfacial damage; 0.355 J cm^{-2} and 0.796 J cm^{-2} seem workable while 1.453 J cm^{-2} and 2.323 J cm^{-2} caused photo darkening. So, for 515 nm wavelength laser (green), high fluence at high pulse width can cause interfacial damage, but at low pulse width, high fluence will cause photo darkening. Following this, lower fluence values for lower pulse widths were tested. For 200 fs and 1000 fs pulse widths, low end of the fluence values possible with this laser setup was insufficient to cause significant

interfacial damage, while higher fluence values caused photodarkening. Since 10,000 fs and 5000 fs pulse widths provided a range of workable fluence values without photodarkening, these values were used for further experiments.

3.2.1. Parameter optimization and assessment of potential damage

To identify the optimal fluence values for 10,000 fs and 5000 fs pulse widths, irradiation experiments with PRR of 100 kHz and 60% spot overlap were carried out in the previously discussed workable fluence range (refer to discussion on Fig. 5(c)). For 10,000 fs and 5000 fs pulse widths, twelve and six fluence values were tested respectively, in the workable range (refer to Table 2). To assess the damage caused by laser treatment, SEM imaging of cell surface, selected area EDS and FTIR analysis of separated EVA are performed. For all the tested fluence values, clean separation was observed between cell and front EVA, so the limiting factors are glass and EVA separation and damage to separated EVA.

Fig. 6(a) shows the comparison of FTIR spectrum of reference EVA and separated EVA of 1.369 J cm^{-2} fluence sample for with 5000 fs pulse

Table 2

Damage summary of the EVA separated by green laser at both 10,000 and 5000 fs pulse widths for different fluence ranges.

Pulse width (fs)	Fluence range (J cm^{-2})	Effect on EVA separation
10,000	<0.355	Low energy, no damage to the interface
10,000	0.355–1.453	Undamaged EVA
10,000	>1.453	Damaged EVA
5000	1.369	Undamaged EVA
5000	Others	Damaged EVA

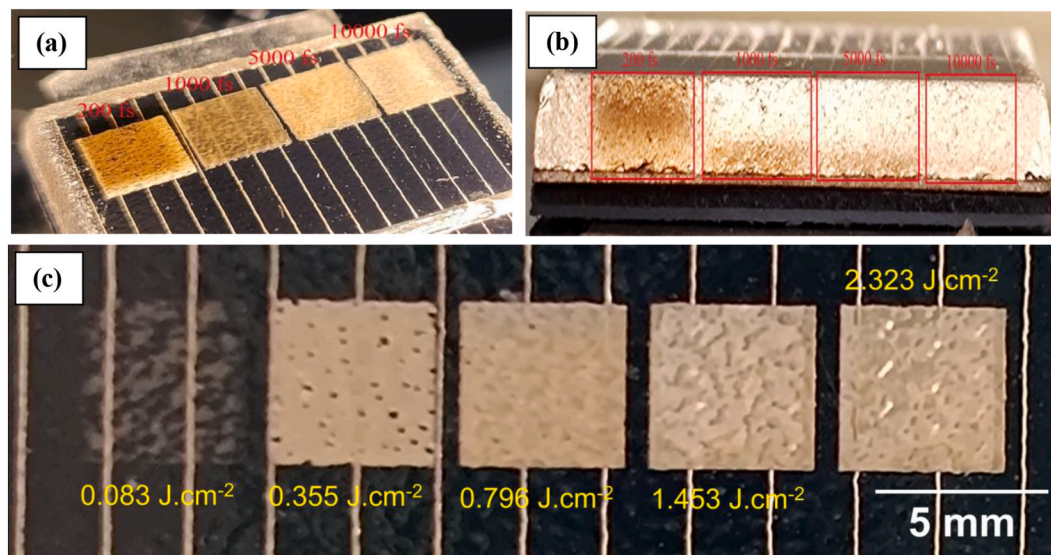


Fig. 5. (a) Four squares irradiated with green laser of pulse widths 200, 1000, 5000 and 10,000 fs from left to right respectively indicating damage at the interface, (b) Corresponding browning of glass and (c) Five squares irradiated with green laser of fluence values 0.083, 0.355, 0.796, 1.453 and 2.323 J cm^{-2} from left to right respectively indicating damage at the interface.

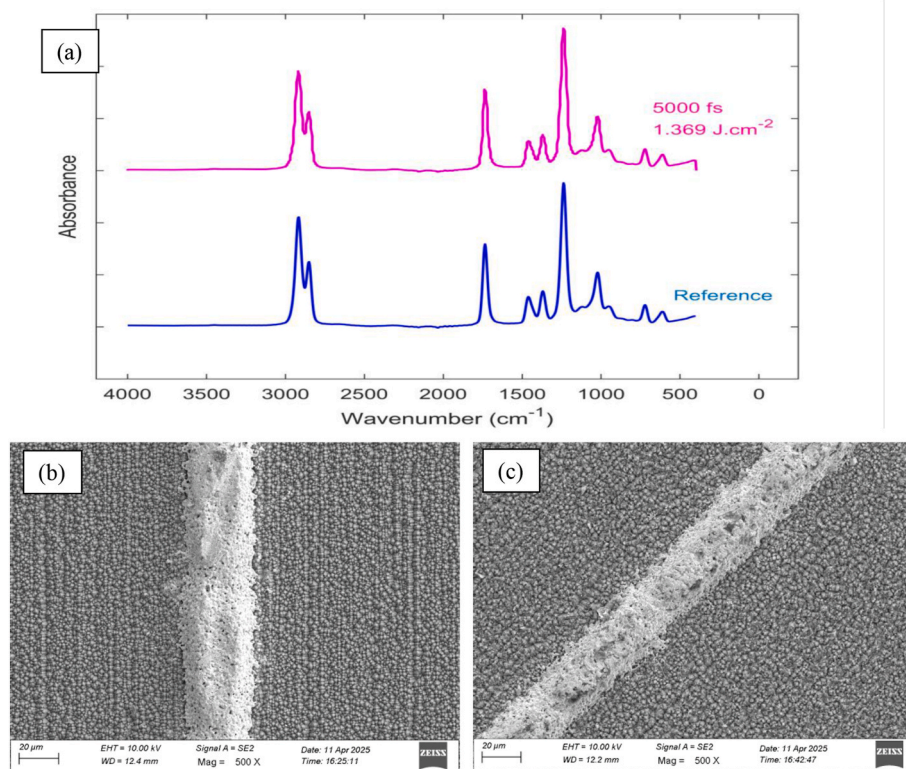


Fig. 6. (a) FTIR of separated EVA – 5000 fs, 1.369 J cm^{-2} , (b) SEM image of reference cell and (c) SEM image of 5000 fs, 1.369 J cm^{-2} sample cell.

width. For 10,000 fs pulse width irradiation, the fluence values between 0.355 J cm^{-2} and 1.453 J cm^{-2} showed no new peaks when compared to reference EVA spectrum. Fluence values lower than this range didn't carry enough energy to damage the interface, while fluence values above this range damaged the separated EVA. For 5000 fs pulse width, only sample irradiated with 1.369 J cm^{-2} fluence value showed no new peaks, i.e., separated EVA is undamaged. Table 2 shows the summary of the optimization data of green laser.

Similar to IR laser treatment, damaged EVA from green laser treatment showed broad peak between 3700 and 3100 cm^{-1} and small peaks between 1700 and 1500 cm^{-1} when compared to reference EVA. Fig. 6 (b) shows the unirradiated cell surface topology SEM image, with expected honeycomb pattern of the antireflective coating. The fluence used on the sample in Fig. 6(c), 1.369 J cm^{-2} , caused surface modification like melting, recrystallization etc., (Jeschke et al., 2002), which is why the image shows that the honeycomb structure is destroyed. Since the top surface is destroyed, the interface bonds are broken, causing the separation of cell and front EVA, along with cell surface turning grey. The selected area EDS analysis of the separated EVA from 5000 fs sample (Fig. 6(c)) shows that there are no metal deposits on the surface.

Interestingly, it was observed that single green laser treatment irradiated with fluence of 0.355 J cm^{-2} , PRR of 100 kHz, 60% spot overlap and 10,000 fs pulse width laser at the front enabled backsheet separation along with glass, front EVA and cell separation. The back EVA and cell were still attached. These results show that green laser has better potential for separation as single treatment separated all the layers except for cell and back EVA. Selected area EDS of back EVA from this sample showed deposits of silicon and aluminum on its surface (also observed in (X. Li et al., 2022)). Further studies are required to optimize the parameters so that all the layers are cleanly separated with a single laser treatment.

In summary, both IR and green laser treatments were effective in achieving the main aim of this study, which is separating the glass, front EVA and the cell. The processing speeds of both lasers are identical when operated at the same pulse repetition rate. Green laser due to conversion

losses, has limited efficiency at higher PRR. For this reason, all the experiments with green laser were conducted at 100 kHz PRR. This is a major disadvantage because processing speed is directly proportional to PRR. The separation quality was identical for both laser treatments. IR lasers offer greater availability and potential applicability than green lasers, because IR technology is well-established and has wider industrial applications.

4. Comparison of the laser delamination process with alternative methods and challenges in upscaling

The proposed method shows promise in PV module delamination, but some challenges remain requiring further research before commercialization. The $30 \text{ mm} \times 25 \text{ mm}$ samples used are much smaller compared to full size solar panel approximately, $2 \text{ m} \times 1 \text{ m}$. The laser setup used in this research is a sophisticated equipment typically used for microfabrication of metals, ceramics etc. As such it is very expensive and not built to be used for recycling solar panels. So, for delamination custom rig with lasers operating at higher PRR (MHz) is required to accelerate the treatment time, complemented by polygon scanner instead of galvanometer scanner to increase the scanning area and reduce the treatment time. Therefore, further research is required before this method can be integrated with the current metal extraction techniques for solar panel recycling. With the system employed in this work, it would take approximately 6 h to delaminate a $2 \text{ m} \times 1 \text{ m}$ commercial solar panel. As mentioned previously, with a custom laser system built for solar panel recycling, the process could become several times faster. In terms of processing time, laser delamination falls between pyrolysis and chemical delamination. The primary advantage of laser method is that it is non-toxic, provides clean separation and produces low waste, while pyrolysis produces toxic gases and chemical methods produce effluents. Laser delamination process will have some wastage, especially at the edges of the panel where cells are not present to focus the laser on. This waste will be reduced considerably when recycling large commercial solar panels, as they are designed for high efficiency and have

minimal non-cellular area at the edges.

5. Summary of discussions

In this article, front EVA layer was cleanly separated from cell and glass through femtosecond laser irradiation treatment. By removing the backsheet first and then irradiating both back and front sides of the module, separation of all the layers of the module was demonstrated to be possible using IR laser. Due to laser ablation, the top surface of the cell got modified disrupting the interfacial bond between cell and front EVA leading to clean separation. The nonlinear nature of the interaction between the laser and glass-EVA interface, along with the heat conducted by the layers when laser is transmitted through them, are the most likely reasons for the breaking of glass-EVA interfacial bond, achieving clean separation between glass and EVA. The separated EVA layers were tested for damage using FTIR and SEM. Selected area EDS analysis was also performed on the irradiated cell and separated EVA layer to look for cell surface damage and metal deposits, respectively. For both 1030 nm (IR) and 515 nm (green) wavelength lasers parameters were optimized to delaminate the module with minimal damage to the EVA layer. Both IR and green lasers were equally effective in delamination, but we observed that IR is more advantageous in terms of processing time, availability and potential applicability. This laser delamination method demonstrates tremendous potential for non-toxic, effective delamination of PV modules, which will help developing an environmentally friendly circular economy for waste solar panels.

6. Conclusion

This work demonstrates that clean separation between glass, front EVA and solar cell can be achieved through femtosecond laser irradiation. By first removing the backsheet and treating both front and back sides of the module with laser, complete layer by layer delamination of module was achieved. For optimized parameters, we found that both 1030 nm and 515 nm lasers effectively delaminated the module with minimum damage to the EVA layer. In addition, we have demonstrated that it is possible to separate glass, front EVA, cell and backsheet with a single 515 nm laser treatment applied to the front side of the module.

CRedit authorship contribution statement

Radhesh Sanathi: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Shantanu Bhowmik:** Writing – review & editing, Supervision. **Sourish Banerjee:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors would like to thank Dr Benjamin Johnston, node manager of Optofab at Macquarie University, for his valuable support and assistance in conducting the experimental work.

Data availability

No data was used for the research described in the article.

References

- Abdo, D.M., Mangialardi, T., Medici, F., Piga, L., 2023. D-Limonene as a promising green solvent for the detachment of end-of-life photovoltaic solar panels under sonication. *Processes* 11 (6). <https://doi.org/10.3390/pr11061848>.
- Agroui, K., Jaunich, M., Arab, A.H., 2016. Analysis techniques of polymeric encapsulant materials for photovoltaic modules: situation and perspectives. *Energy Proc.* 93, 203–210. <https://doi.org/10.1016/j.egypro.2016.07.171>.
- Agroui, K., Maallem, A., Boumaour, M., Collins, G., Salama, M., 2006. Thermal stability of slow and fast cure EVA encapsulant material for photovoltaic module manufacturing process. *Sol. Energy Mater. Sol. Cell.* 90 (15), 2509–2514. <https://doi.org/10.1016/j.solmat.2006.03.023>.
- Al Zaabi, B., Ghosh, A., 2024. Managing photovoltaic waste: sustainable solutions and global challenges. *Sol. Energy* 283, 112985. <https://doi.org/10.1016/j.solener.2024.112985>.
- Azizi, S., David, E., Fréchet, M.F., Nguyen-Tri, P., Ouellet-Plamondon, C.M., 2018. Electrical and thermal conductivity of ethylene vinyl acetate composite with graphene and carbon black filler. *Polym. Test.* 72, 24–31. <https://doi.org/10.1016/j.polymertesting.2018.09.031>.
- Bandegi, A., Gray, T.G., Mitchell, S., Jamei Oskouei, A., Sing, M.K., Kennedy, J., Miller McLoughlin, K., Manas-Zloczower, I., 2023. Vitrimization of crosslinked elastomers: a mechanochemical approach for recycling thermoset polymers [10.1039/D3MA00098B]. *Mater. Adv.* 4 (12), 2648–2658. <https://doi.org/10.1039/D3MA00098B>.
- Barretta, C., Oreski, G., Feldbacher, S., Resch-Fauster, K., Pantani, R., 2021. Comparison of degradation behavior of newly developed encapsulation materials for photovoltaic applications under different artificial ageing tests. *Polymers* 13 (2).
- Bin Anwar, T., Hanson, K.M., Lam, K., Bardeen, C.J., 2024. Using nanosecond laser pulses to debond the glass-EVA layer from silicon photovoltaic modules. *Waste Manag.* 187, 275–284. <https://doi.org/10.1016/j.wasman.2024.07.013>.
- Brand, J., Wain, A., Rode, A.V., Madden, S., King, P.L., Rapp, L., 2023. Comparison between nanosecond and femtosecond laser pulses for the removal of spray paint from granite surfaces. *J. Cult. Herit.* 62, 329–338. <https://doi.org/10.1016/j.culher.2023.06.005>.
- Chen, H., Li, Z., Han, M., Yang, X., Bai, S., 2024. Improved femtosecond laser welding of nonoptical contact glass by pressure and a water intermediate layer. *ACS Omega* 9 (37), 38878–38886. <https://doi.org/10.1021/acsomega.4c05018>.
- Chitra, Sah, D., Lodhi, K., Kant, C., Saini, P., Kumar, S., 2020. Structural composition and thermal stability of extracted EVA from silicon solar modules waste. *Sol. Energy* 211, 74–81. <https://doi.org/10.1016/j.solener.2020.09.039>.
- Chris, B.S., André, B., Eric, M., 2001. Laser-induced breakdown and damage in bulk transparent materials induced by tightly focused femtosecond laser pulses. *Meas. Sci. Technol.* 12 (11), 1784. <https://doi.org/10.1088/0957-0233/12/11/305>.
- Dias, P., Javimczik, S., Benevit, M., Veit, H., 2017. Recycling WEEE: polymer characterization and pyrolysis study for waste of crystalline silicon photovoltaic modules. *Waste Manag.* 60, 716–722. <https://doi.org/10.1016/j.wasman.2016.08.036>.
- Dobra, T., Florian, T., Gerhard, W., Daniel, V., Pomberger, R., 2023. Selective delamination by milling as a first step in the recycling of photovoltaic modules. *Environ. Technol.* 44 (22), 3437–3445. <https://doi.org/10.1080/09593330.2022.2061380>.
- Doi, T., Tsuda, I., Unagida, H., Murata, A., Sakuta, K., Kurokawa, K., 2001. Experimental Study on PV Module Recycling with Organic Solvent Method (Solar Energy Materials & Solar Cells, Issue).
- Fiandra, V., Sannino, L., Andreozzi, C., 2023. Photovoltaic waste as source of valuable materials: a new recovery mechanical approach. *J. Clean. Prod.* 385. <https://doi.org/10.1016/j.jclepro.2022.135702>.
- Fiandra, V., Sannino, L., Andreozzi, C., Corcelli, F., Graditi, G., 2019a. Silicon photovoltaic modules at end-of-life: removal of polymeric layers and separation of materials. *Waste Manag.* 87, 97–107. <https://doi.org/10.1016/j.wasman.2019.02.004>.
- Fiandra, V., Sannino, L., Andreozzi, C., Graditi, G., 2019b. End-of-life of silicon PV panels: a sustainable materials recovery process. *Waste Manag.* 84, 91–101. <https://doi.org/10.1016/j.wasman.2018.11.035>.
- Granata, G., Pagnanelli, F., Moscardini, E., Havlik, T., Toro, L., 2014. Recycling of photovoltaic panels by physical operations. *Sol. Energy Mater. Sol. Cell.* 123, 239–248. <https://doi.org/10.1016/j.solmat.2014.01.012>.
- Guo, B., Sun, J., Lu, Y., Jiang, L., 2019. Ultrafast dynamics observation during femtosecond laser-material interaction. *Int. J. Extrem. Manuf.* 1 (3), 032004. <https://doi.org/10.1088/2631-7990/ab3a24>.
- Hallam, B., Kim, M., Zhang, Y., Wang, L., Lennon, A., Verlinden, P., Altermatt, P.P., Dias, P.R., 2023. The silver learning curve for photovoltaics and projected silver demand for net-zero emissions by 2050. *Prog. Photovoltaics Res. Appl.* 31 (6), 598–606. <https://doi.org/10.1002/ppp.3661>.
- IEA-PVPS, I. a., 2016. End-of-Life management: solar photovoltaic panels. I. R. E. A. a. I. E. A. P. P. Systems. <https://www.irena.org/publications/2016/Jun/End-of-life-management-Solar-Photovoltaic-Panels>.
- IEA, 2025. Electricity 2025. <https://www.iea.org/reports/electricity-2025>.
- Ito, M., Doi, T., 2024. Recycling technology using hot knife separation method. In: 2024 31st International Workshop on Active-Matrix Flatpanel Displays and Devices (AM-FPD), 2–5 July 2024.
- Jain, S., Sharma, T., Gupta, A.K., 2022. End-of-life management of solar PV waste in India: situation analysis and proposed policy framework. *Renew. Sustain. Energy Rev.* 153, 111774. <https://doi.org/10.1016/j.rser.2021.111774>.
- Jeschke, H.O., Garcia, M.E., Lenzner, M., Bonse, J., Krüger, J., Kautek, W., 2002. Laser ablation thresholds of silicon for different pulse durations: theory and experiment.

- Appl. Surf. Sci. 197–198, 839–844. [https://doi.org/10.1016/S0169-4332\(02\)00458-0](https://doi.org/10.1016/S0169-4332(02)00458-0).
- Keiichi Komoto, C.A., Alonso-Garcia, Carmen, Costa, Daniele, Curtis, Taylor, Danelli, Andrea, Deng, Rong, Defrenne, Nicolas, Drozdziak, Karen, Ebin, Burcak, Espinosa, Nieves, Gassner, Anika, Heath, Garvin, Held, Michael, Lee, Jin-Seok, Libby, Cara, Lyu, Fang, Mirlletz, Heather, Penn, Michaela, Shen, Yansong, Smith, Ligia, Stucki, Matthias, Tammara, Marco, Theelen, Mirjam, Vogt, Malte R., Wessendorf, Cordula, Woods-Robinson, Rachel, 2025. Status of PV Module Recycling in IEA PVPS Task 12 Countries.
- Kim, J., 2011. Optimization of SiNx layer for solar cell using computational method. *Curr. Appl. Phys.* 11 (1, Suppl. ment), S39–S42. <https://doi.org/10.1016/j.cap.2010.11.048>.
- Kim, Y., Lee, J., 2012. Dissolution of ethylene vinyl acetate in crystalline silicon PV modules using ultrasonic irradiation and organic solvent. *Sol. Energy Mater. Sol. Cell.* 98, 317–322. <https://doi.org/10.1016/j.solmat.2011.11.022>.
- Koenig, J.L., Boerio, F.J., Plueddemann, E.P., Miller, J., Willis, P.B., Cuddihy, E.F., 1986. *Chemical Bonding Technology: Direct Investigation of Interfacial Bonds*.
- Kwak, J.I., Nam, S.-H., Kim, L., An, Y.-J., 2020. Potential environmental risk of solar cells: current knowledge and future challenges. *J. Hazard Mater.* 392, 122297. <https://doi.org/10.1016/j.jhazmat.2020.122297>.
- Lee, J.K., Lee, J.S., Ahn, Y.S., Kang, G.H., Song, H.E., Kang, M.G., Kim, Y.H., Cho, C.H., 2018. Simple pretreatment processes for successful reclamation and remanufacturing of crystalline silicon solar cells. *Prog. Photovoltaics Res. Appl.* 26 (3), 179–187. <https://doi.org/10.1002/pip.2963>.
- Lenzner, M., Krüger, J., Sartania, S., Cheng, Z., Spielmann, C., Mourou, G., Kautek, W., Krausz, F., 1998. Femtosecond optical breakdown in dielectrics. *Phys. Rev. Lett.* 80 (18), 4076–4079. <https://doi.org/10.1103/PhysRevLett.80.4076>.
- Li, K., Wang, Z., Liu, C., Wang, D., Li, G., Chen, X., Qian, G., Hu, K., 2022. A green method to separate different layers in photovoltaic modules by using DMPU as a separation agent. *Sol. Energy Mater. Sol. Cell.* 245. <https://doi.org/10.1016/j.solmat.2022.111870>, 111870–111870.
- Li, X., Liu, H., You, J., Diao, H., Zhao, L., Wang, W., 2022. Back EVA recycling from c-Si photovoltaic module without damaging solar cell via laser irradiation followed by mechanical peeling. *Waste Manag.* 137, 312–318. <https://doi.org/10.1016/j.wasman.2021.11.024>.
- Liu, Y.-H., Cheng, C.-W., 2023. The experimental and modeling study of femtosecond laser-ablated silicon surface. *J. Manuf. Mater. Process.* 7 (2).
- Macku, R., Koktavy, P., Šicner, J., 2012. Comprehensive study of solar cell structure defects by means of noise and light emission analysis. *Adv. Electr. Electron. Eng.* 10. <https://doi.org/10.15598/aeec.v10i2.620>.
- Meena, R., Pareek, A., Gupta, R., 2024. A comprehensive review on interfacial delamination in photovoltaic modules. *Renew. Sustain. Energy Rev.* 189, 113944. <https://doi.org/10.1016/j.rser.2023.113944>.
- Min, R., Li, K., Wang, D., Xiao, W., Liu, C., Wang, Z., Bian, S., 2023. A novel method for layer separation of photovoltaic modules by using green reagent EGDA. *Sol. Energy* 253, 117–126. <https://doi.org/10.1016/j.solener.2023.02.024>.
- Ngagoum Ndalloka, Z., Vijayakumar Nair, H., Alpert, S., Schmid, C., 2024. Solar photovoltaic recycling strategies. *Sol. Energy* 270, 112379. <https://doi.org/10.1016/j.solener.2024.112379>.
- Oreski, G., Wallner, G.M., Lang, R.W., 2009. Ageing characterization of commercial ethylene copolymer greenhouse films by analytical and mechanical methods. *Biosyst. Eng.* 103 (4), 489–496. <https://doi.org/10.1016/j.biosystemseng.2009.05.003>.
- Pagnanelli, F., Moscardini, E., Granata, G., Abo Atia, T., Altimari, P., Havlik, T., Toro, L., 2017. Physical and chemical treatment of end of life panels: an integrated automatic approach viable for different photovoltaic technologies. *Waste Manag.* 59, 422–431. <https://doi.org/10.1016/j.wasman.2016.11.011>.
- Pan, X., Wang, Y., Bai, H., Ren, C., Zhang, W., Wang, J., Peng, J., Jing, F., Qiu, H., Zhu, C., Liu, H., Yang, C., Hu, Z., Wu, Y., 2023. Ultrafast damage dynamics and ablation mechanism of GaSe induced by femtosecond laser irradiation. *Appl. Surf. Sci.* 637, 157999. <https://doi.org/10.1016/j.apsusc.2023.157999>.
- Säckl, G., Wallner, G.M., Duchoslav, J., Tiefenthaler, M., Stifter, D., 2024. Advanced analysis of ethylene vinyl acetate copolymer materials for photovoltaic modules. *Polym. Test.* 132, 108381. <https://doi.org/10.1016/j.polymertesting.2024.108381>.
- Sharma, A., Slipchenko, M.N., Shneider, M.N., Wang, X., Rahman, K.A., Shashurin, A., 2018. Counting the electrons in a multiphoton ionization by elastic scattering of microwaves. *Sci. Rep.* 8 (1), 2874. <https://doi.org/10.1038/s41598-018-21234-y>.
- Sikora, A., Grojo, D., Sentis, M., 2017. Wavelength scaling of silicon laser ablation in picosecond regime. *J. Appl. Phys.* 122 (4), 045702. <https://doi.org/10.1063/1.4994307>.
- Song, B.P., Zhang, M.Y., Fan, Y., Jiang, L., Kang, J., Gou, T.T., Zhang, C.L., Yang, N., Zhang, G.J., Zhou, X., 2020. Recycling experimental investigation on end of life photovoltaic panels by application of high voltage fragmentation. *Waste Manag.* 101, 180–187. <https://doi.org/10.1016/j.wasman.2019.10.015>.
- Spectroscopy, 2013. Femtosecond vs. nanosecond laser pulse duration for laser ablation chemical analysis. *Spectroscopy*. <https://www.spectroscopyonline.com/view/femtosecond-vs-nanosecond-laser-pulse-duration-laser-ablation-chemical-analysis>.
- Technology, K., 2025. Plasma surface modification: a breakthrough for solar energy systems and new energy. <https://www.keylinktech.com/plasma-surface-technology/application/solar-energy/>.
- Tembo, P.M., Heninger, M., Subramanian, V., 2021. An investigation of the recovery of silicon photovoltaic cells by application of an organic solvent method. *ECS J. Solid State Sci. Technol.* 10 (2). <https://doi.org/10.1149/2162-8777/abe093>, 025001–025001.
- Uličná, S., Sinha, A., Miller, D.C., Habersberger, B.M., Schelhas, L.T., Owen-Bellini, M., 2023. PV encapsulant formulations and stress test conditions influence dominant degradation mechanisms. *Sol. Energy Mater. Sol. Cell.* 255, 112319. <https://doi.org/10.1016/j.solmat.2023.112319>.
- Volk, A.-K., Tucher, N., Seiffe, J., Hauser, H., Zimmer, M., Bläsi, B., Hofmann, M., Rentsch, J., 2015. Honeycomb structure on multi-crystalline silicon Al-BSF solar cell with 17.8% efficiency. *IEEE J. Photovoltaics* 5 (4), 1027–1033.
- Wahman, M., Surowiak, A., Ebin, B., Berent, K., 2024. PV back sheet recovery from c-Si modules using hot knife technique. *Sol. Energy Mater. Sol. Cell.* 276, 113067. <https://doi.org/10.1016/j.solmat.2024.113067>.
- Weber, R., Graf, T., Berger, P., Onuseit, V., Wiedenmann, M., Freitag, C., Feuer, A., 2014. Heat accumulation during pulsed laser materials processing. *Opt. Express* 22 (9), 11312–11324. <https://doi.org/10.1364/OE.22.011312>.
- Wright, E.M., Koch, S.W., Kolesik, M., Moloney, J.V., 2019. Memory effects in the long-wave infrared avalanche ionization of gases: a review of recent progress. *Rep. Prog. Phys.* 82 (6), 064401. <https://doi.org/10.1088/1361-6633/ab1a07>.
- Xu, X., Lai, D., Wang, G., Wang, Y., 2021. Nondestructive silicon wafer recovery by a novel method of solvothermal swelling coupled with thermal decomposition. *Chem. Eng. J.* 418. <https://doi.org/10.1016/j.cej.2021.129457>.
- Xue, H., Xu, Z., Zhang, M., Wang, J., Ruan, W., 2014. Ethylene vinyl acetate films filled with ytterbium containing rare earth particles (Y2SiO5: Ce³⁺, Yb³⁺) which have optical down-conversion capabilities and useful for encapsulating solar cells. *J. Plast. Film Sheeting* 31 (3), 233–247. <https://doi.org/10.1177/8756087914540673>.
- Yan, Y., Wang, Z., Wang, D., Cao, J., Ma, W., Wei, K., Yun, L., 2020. Recovery of silicon via using KOH-ethanol solution by separating different layers of end-of-life PV modules. *JOM* 72 (7), 2624–2632. <https://doi.org/10.1007/s11837-020-04193-6>.
- Yang, C., Ji, C., Feng, S., Liu, Y., Wei, W., Long, Y., 2025. Ultrafast laser-matter interaction mechanisms and applications in functional device fabrication: recent advances and perspectives. *Appl. Phys. Rev.* 12 (3), 031325. <https://doi.org/10.1063/5.0228383>.
- Yu, Y., Li, S., Lu, J., Xi, F., Chen, X., Wu, D., Ma, W., Deng, R., 2024. Green recycling of end-of-life photovoltaic modules via deep-eutectic solvents. *Chem. Eng. J.* 499, 155933. <https://doi.org/10.1016/j.cej.2024.155933>.